TECHNICAL BRIEF · WEEK 4 (REVISED)

Why We Must Automate Our Data Pipelines

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For: Senior Operations Management
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Every week, someone on our team manually re-types depot stock counts, sales numbers and shipment records into a spreadsheet. It works until it doesn't. This brief covers the risk, the fix, how the automated pipeline runs, and what it is worth to us.

1. The Risk: One Typo, One Bad Week

A warehouse clerk closes out Friday's count. Depot C6 really holds **1,240** units, but the spreadsheet says **12,400**. One extra digit, and nothing checks it. On Monday the replenishment plan sees an over-stocked depot and skips it, while the real shelf is nearly empty. By midweek C6 is out of stock, deliveries slip and an order is missed. This is not rare: it happens whenever a person re-types numbers by hand.

The uncomfortable truth: Excel cannot tell a real number from a typo. It trusts whatever we type and passes the mistake into every decision built on top of it.

2. The Solution: Automatic Irrigation, Not Buckets

Today's process is watering a field by carrying buckets from a well: it depends on someone remembering to go and not spilling any. An *automated pipeline* is a sprinkler system: data flows from the source, is checked, and is delivered on a schedule with no one re-typing it.

TODAY: BUCKETS
Manual re-typing, one field at a time. A tired hand or a missed row silently breaks the plan.

AUTOMATED: SPRINKLERS
Built-in checks (*data validation*) catch typos and gaps before they reach a decision.

3. How It Runs: Safe, Scheduled and Visible

- **Runs by itself.** A scheduled job starts the pipeline every night at 2:00 AM. No one has to remember.
- **Stops on bad data.** A quality gate (Great Expectations plus business rules) checks the data before forecasting or loading. If it fails, the run halts and nothing reaches the database.
- **Honest logging.** Every stage records its start time, end time and duration. A failed run is logged as failed and exits with an error code, so it is never reported as a success.
- **Safe to re-run.** The database load replaces tables rather than appending, so a repeat run cannot duplicate rows.
- **Clear ownership.** On a failed run, the operations analyst reviews the log and re-runs after the data issue is fixed.

4. The Value: What It Is Worth to Us

In a test run against deliberately simulated bad data, the automated checks caught **over 4,300 inventory mismatches** and flagged **13,000+ records** where a depot was at risk of exceeding capacity or running out of stock, all before any decision used those numbers. A spreadsheet catches none of this on its own. Fewer hours go on re-typing and reconciling, and stock-out warnings arrive the same day instead of a week later.

4,300+	13,000+	2:00 AM
mismatches caught in test run before reaching a report	capacity / stock-out risk records flagged	automatic nightly run, stops itself on bad data

Bottom line: Excel can't check itself. This pipeline checks every number, runs every night, and stops itself when something looks wrong.